

Math 344, F18
Final Exam

Name: *Answer*

Instructions:

- There are totally 7 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	12	
2	15	
3	16	
4	14	
5	15	
6	14	
7	14	
Total	100	

Good Luck !

Problem 1. (12 points) Given two vectors in the three dimensional space $\mathbf{v} = (1, \frac{\sqrt{3}}{3}, \frac{2\sqrt{6}}{3})$ and $\mathbf{w} = (0, -\frac{2\sqrt{6}}{3}, -\frac{2\sqrt{3}}{3})$.

(a) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{1^2 + (\frac{\sqrt{3}}{3})^2 + (\frac{2\sqrt{6}}{3})^2} = \sqrt{1 + \frac{1}{3} + \frac{8}{3}} = \sqrt{4} = 2$$

$$\|\mathbf{w}\| = \sqrt{0^2 + (-\frac{2\sqrt{6}}{3})^2 + (-\frac{2\sqrt{3}}{3})^2} = \sqrt{0 + \frac{8}{3} + \frac{4}{3}} = \sqrt{4} = 2$$

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= 1 \cdot 0 + \frac{\sqrt{3}}{3} \cdot (-\frac{2\sqrt{6}}{3}) + \frac{2\sqrt{6}}{3} \cdot (-\frac{2\sqrt{3}}{3}) \\ &= -\frac{2\sqrt{2}}{3} - \frac{4\sqrt{2}}{3} = -2\sqrt{2} \end{aligned}$$

$$\Rightarrow \cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{-2\sqrt{2}}{4} = -\frac{\sqrt{2}}{2}$$

$$\Rightarrow \theta = \frac{3}{4}\pi$$

(b) Choose a number c so that $2\mathbf{w} + c\mathbf{v}$ is perpendicular to $2\mathbf{w} - c\mathbf{v}$.

$$(2\mathbf{w} + c\mathbf{v})(2\mathbf{w} - c\mathbf{v}) = 0$$

$$4\|\mathbf{w}\|^2 - c^2\|\mathbf{v}\|^2 = 0$$

$$c^2 = \frac{4\|\mathbf{w}\|^2}{\|\mathbf{v}\|^2}$$

$$= 4$$

$$\Rightarrow c = \pm 2$$

Problem 2. (15 points) Let $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 2 & 1 \\ 2 & 8 & 6 \end{bmatrix}$.

(a) Find a row exchange matrix P and B such that the Gaussian elimination can work on the matrix $B = PA$.

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad B = PA = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \\ 2 & 8 & 6 \end{bmatrix}$$

$$(\text{or } P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix})$$

(b) Find B^{-1} using the Gaussian-Jordan Elimination.

$$\begin{aligned} [B|I] &= \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 2 & 8 & 6 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 4 & 4 & -2 & 0 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 2 & -2 & -2 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -1 & -1 & \frac{1}{2} \end{array} \right] = [I|B^{-1}] \\ &\Rightarrow B^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & -1 & \frac{1}{2} \end{bmatrix} \end{aligned}$$

(c) Find A^{-1} based on results from (a) and (b).

$$\begin{aligned} A^{-1} &= (PB)^{-1} = B^{-1}P^{-1} = B^{-1}P \quad \leftarrow \text{exchange columns 1 \& 2.} \\ &= \begin{bmatrix} -1 & \frac{1}{2} & 0 \\ 1 & \frac{1}{2} & -\frac{1}{4} \\ -1 & -1 & \frac{1}{2} \end{bmatrix} \end{aligned}$$

Problem 3. (16 points) Let $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 2 & -8 \end{bmatrix}$.

(a) Find R , the reduced echelon form of A .

$$A = \begin{bmatrix} \textcircled{1} & 2 & 1 & 0 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 2 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & \textcircled{2} & 8 \\ 0 & 0 & -2 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & \textcircled{2} & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \rightarrow \begin{bmatrix} 1 & 2 & 0 & -4 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 0 & -4 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(R)) = 2 \quad \leftarrow \text{columns 1 \& 3 of } A \\ \text{Basis: } \left\{ \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\} \quad (\text{not columns 1 \& 3 of } R)$$

(c) Find the dimension and a basis for the nullspace of A , $N(A)$.

Pivot variables x_1, x_3 ; free variables x_2, x_4

$$\dim(N(A)) = 4 - 2 = 2$$

Solve $Rx = 0$

$$\begin{cases} x_1 + 2x_2 - 4x_4 = 0 \\ x_3 + 4x_4 = 0 \end{cases}$$

$$\text{Set } x_2 = 1, x_4 = 0 \Rightarrow s_1 = (-2, 1, 0, 0)$$

$$\text{Set } x_2 = 0, x_4 = 1 \Rightarrow s_2 = (4, 0, -4, 1)$$

Thus Basis :

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ -4 \\ 1 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$\dim(N(A^T)) = 3 - 2 = 1$$

$$\text{Solve } A^T y = 0 \Rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 4 & 8 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} y_1 + 2y_2 + 4y_3 = 0 \\ 2y_2 - y_3 = 0 \end{cases} \Rightarrow \begin{cases} y_1 = -6y_3 \\ y_2 = y_3 \end{cases} \Rightarrow y = c \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{Basis: } \left\{ \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Problem 4. (14 points)

(a) Find the linear combination of $\mathbf{a}_1 = (1, 2, -1)$ and $\mathbf{a}_2 = (1, 1, 1)$ that is closest to $\mathbf{b} = (2, 1, 1)$.

$$A\mathbf{x} = \mathbf{b} \text{ where } A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{Solve } A^T A \hat{\mathbf{x}} = A^T \mathbf{b} \rightarrow \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\Rightarrow x_1 = \frac{1}{14} \quad x_2 = \frac{9}{7}$$

$$\Rightarrow A \hat{\mathbf{x}} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{14} \\ \frac{9}{7} \end{bmatrix} = \begin{bmatrix} \frac{19}{14} \\ \frac{10}{7} \\ \frac{17}{14} \end{bmatrix}$$

(b) Find two orthonormal vectors $\mathbf{q}_1$ and $\mathbf{q}_2$ in the plane spanned by $\mathbf{a}_1 = (1, 3, 4, 5, 7)$ and $\mathbf{a}_2 = (-6, 6, 8, 0, 8)$ by the Gram-Schmidt process.

$$\tilde{\mathbf{q}}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \\ 7 \end{bmatrix} \quad \|\tilde{\mathbf{q}}_1\| = \sqrt{1^2 + 3^2 + 4^2 + 5^2 + 7^2} = \sqrt{100} = 10$$

$$\Rightarrow \mathbf{q}_1 = \frac{1}{10} \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1/10 \\ 3/10 \\ 2/5 \\ 1/2 \\ 7/10 \end{bmatrix}$$

$$\tilde{\mathbf{q}}_2 = \mathbf{a}_2 - (\mathbf{a}_2^T \mathbf{q}_1) \mathbf{q}_1 = \begin{bmatrix} -6 \\ 6 \\ 8 \\ 0 \\ 8 \end{bmatrix} - \frac{100}{10} \begin{bmatrix} 1/10 \\ 3/10 \\ 2/5 \\ 1/2 \\ 7/10 \end{bmatrix} = \begin{bmatrix} -7 \\ 3 \\ 4 \\ -5 \\ 1 \end{bmatrix}$$

$$\|\tilde{\mathbf{q}}_2\| = 10 \Rightarrow \mathbf{q}_2 = \frac{1}{10} \begin{bmatrix} -7 \\ 3 \\ 4 \\ -5 \\ 1 \end{bmatrix} = \begin{bmatrix} -7/10 \\ 3/10 \\ 2/5 \\ -1/2 \\ 1/10 \end{bmatrix}$$

Problem 5. (15 points)

(a) Construct a matrix $\mathbf{A}$ such that its column space $C(\mathbf{A})$ contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and its row space $C(\mathbf{A}^T)$ contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$$\text{Just } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Since } \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

(b) Does it always hold $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} e-d & e & f \\ b-a & b & c \\ h-g & h & i \end{vmatrix}$? Please justify your answer.

Yes.

$$\begin{vmatrix} e-d & e & f \\ b-a & b & c \\ h-g & h & i \end{vmatrix} = \begin{vmatrix} -d & e & f \\ -a & b & c \\ -g & h & i \end{vmatrix} = - \begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

(c) Is the inverse of a symmetric matrix also symmetric if it exists? Please justify your answer.

Yes.

When $\mathbf{A} = \mathbf{A}^T$ and $\mathbf{A}^{-1}$ exists, we have that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \Rightarrow (\mathbf{A}\mathbf{A}^{-1})^T = \mathbf{I} \Rightarrow (\mathbf{A}^{-1})^T \mathbf{A}^T = \mathbf{I} \Rightarrow (\mathbf{A}^{-1})^T \mathbf{A} = \mathbf{I}$$

$$\Leftrightarrow \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}$$

$$\Rightarrow ((\mathbf{A}^{-1})^T - \mathbf{A}^{-1}) \mathbf{A} = \mathbf{0} \Rightarrow (\mathbf{A}^{-1})^T = \mathbf{A}^{-1}$$

Problem 6. (14 points) Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 0 & 3 & -2 \\ 0 & -4 & 1 \end{bmatrix}.$$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -1 & 2 \\ 0 & 3-\lambda & -2 \\ -4 & 1-\lambda & \end{vmatrix} = (2-\lambda)[(3-\lambda)(1-\lambda)-8]$$

$$= (2-\lambda)(\lambda^2 - 4\lambda + 3 - 8)$$

$$= -(\lambda-2)(\lambda^2 - 4\lambda - 5) = -(\lambda-2)(\lambda-5)(\lambda+1)$$

$$\Rightarrow \lambda_1 = 2, \lambda_2 = 5, \lambda_3 = -1$$

For $\lambda_1 = 2$, we solve $(A - 2I)X_1 = 0$

$$\begin{bmatrix} 0 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \rightarrow \begin{bmatrix} 0 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} -x_2 + 2x_3 = 0 \\ -9x_3 = 0 \end{cases} \Rightarrow \underline{X_1 = c \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}$$

$\hookrightarrow x_2 = x_3 = 0$

For $\lambda_2 = 5$, we solve $(A - 5I)X_2 = 0$

$$\begin{bmatrix} -3 & -1 & 2 \\ 0 & -2 & -2 \\ 0 & -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \rightarrow \begin{bmatrix} -3 & -1 & 2 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} -3x_1 - x_2 + 2x_3 = 0 \\ -2x_2 - 2x_3 = 0 \end{cases}$$

set $x_3 = 1$, we get $x_2 = -1, x_1 = \frac{1}{3} \Rightarrow \underline{X_2 = c \begin{bmatrix} 1/3 \\ -1 \\ 1 \end{bmatrix}}$

For $\lambda_3 = -1$, we solve $(A + I)X_3 = 0$

$$\begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & -2 \\ 0 & -4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \rightarrow \begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} 3x_1 - x_2 + 2x_3 = 0 \\ 4x_2 - 2x_3 = 0 \end{cases}$$

set $x_3 = 1$, we get $x_2 = \frac{1}{2}, x_1 = -\frac{1}{2} \Rightarrow \underline{X_3 = c \begin{bmatrix} -1/2 \\ 1/2 \\ 1 \end{bmatrix}}$

Problem 7. (14 points)

(a) Find A^{-1} using the cofactor formula $A^{-1} = \frac{C^T}{\det A}$, where $A = \begin{bmatrix} 1 & -2 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

$$\det A = 1 \cdot \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} -2 & 0 \\ -1 & 2 \end{vmatrix} = 3 - 4 = -1$$

$$C = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{C^T}{\det A} = \frac{1}{(-1)} \begin{bmatrix} 3 & 4 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -4 & -2 \\ -2 & -2 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

(b) Let $A = \begin{bmatrix} 2 & -1 & 2b \\ -1 & 2 & -1 \\ 2b & -1 & 2 \end{bmatrix}$. Compute $\det(A)$ and determine for which number b such that $\det(A) > 0$. Is then A is symmetric positive definite in such case?

$$\det A = 2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 \\ 2b & 2 \end{vmatrix} + (2b) \begin{vmatrix} -1 & 2 \\ 2b & -1 \end{vmatrix}$$

$$= 2 \cdot 3 + (-2 + 2b) + (2b)(1 - 4b)$$

$$= 6 - 2 + 2b + 2b - 8b^2 = -8b^2 + 4b + 4$$

$$\det A > 0 \Rightarrow -8b^2 + 4b + 4 > 0 \Rightarrow -4(2b^2 - b - 1) > 0$$

$$\Rightarrow 2b^2 - b - 1 < 0 \Rightarrow (b-1)(2b+1) < 0$$

$$\Rightarrow -\frac{1}{2} < b < 1$$

Yes, in this case, A is positive definite since

$$\det(2) > 0, \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 > 0, \det A > 0.$$